

Final Year Project Report

A Starlink-Like LEO Multi-Hop Routing Simulator and Empirical Evaluation of the BPP Analytical Model

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Reliability Analysis of Multi-hop Routing in Multi-tier LEO Satellite Networks.
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Abstract

This project builds a Starlink-like multi-hop LEO routing simulator from real Two-Line Element (TLE) data. Starting from publicly available Starlink TLEs, the simulator generates ten frozen-time network snapshots covering 81 minutes of orbital movement, each containing 4,080 satellites and 250 ground gateways linked by roughly 37,000 inter-satellite and gateway-access edges.

Two routing algorithms are implemented and compared: Dijkstra shortest-path routing and the greedy Wang-style BPP algorithm from Wang, Kishk, and Alouini (2023) [1]. Both are run on the same real orbital geometry, and their empirical interruption probabilities are compared against the paper’s closed-form BPP analytical prediction.

At $d_{th} = 4,000$ km on the full constellation, BPP predicts 1.44% interruption; the empirical greedy simulator measures 78.57%—an absolute error of 77 percentage points. Dijkstra achieves zero interruptions on every tested pair, confirming that the Walker-delta topology is globally routable and that greedy routing failures are algorithmic, not structural.

A full-featured 3D browser visualiser built on Three.js accompanies the simulator, showing real satellite positions, ISL links, animated packets, and a live traffic simulation with congestion heatmaps, M/D/1 queuing, and a performance dashboard.

Keywords: LEO, Starlink, multi-hop routing, BPP, Walker-delta, TLE, Dijkstra, interruption probability, M/D/1 queueing, 3D visualisation.

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Chapter 1

Introduction

LEO satellite constellations such as Starlink, Kuiper, and OneWeb are deploying tens of thousands of satellites at 340–600 km altitude to deliver sub-50 ms broadband globally. Unlike geostationary satellites ($\sim 35,786$ km, ~ 600 ms RTT), LEO satellites can achieve competitive latency for interactive applications.

Because real Starlink routing tables and traffic data are proprietary, simulation from public TLE data is the main academic tool for studying routing behaviour. Analytical models such as the BPP model of Wang et al. offer closed-form predictions without simulation, but they assume satellites are placed randomly—a strong simplification relative to real Walker-delta orbital geometry.

This project asks: *does the BPP analytical prediction hold when tested against real Starlink TLE-derived geometry?* To answer this, we build a complete simulator that runs both the paper’s own greedy routing algorithm and a Dijkstra baseline on the same real topology, then compare the results against the closed-form BPP prediction.

Chapter 2

Problem Statement and Objectives

2.1 Problem Statement

Three sub-problems motivate this project:

1. **Data gap.** Real Starlink routing behaviour is proprietary; TLE data provides positions but no routing information.
2. **Model validity.** The BPP formula assumes randomly placed satellites. Real Starlink follows Walker-delta orbital geometry, which is not random.
3. **Algorithm suitability.** The paper’s greedy routing algorithm is designed for BPP geometry. Whether it succeeds on a real Starlink-like topology has not been empirically tested.

2.2 Objectives

- Build a Starlink-like LEO routing simulator from real TLE data.
- Generate multi-epoch network snapshots (satellites + gateways + ISL/access edges).
- Implement the Wang/BPP greedy routing algorithm exactly as specified in [1].
- Implement Dijkstra shortest-delay routing as a global-knowledge baseline.
- Compare both empirical results against the BPP closed-form analytical prediction.
- Measure routing success rate, hop count, end-to-end latency, and delay distribution.
- Build a real-time 3D visualiser with live traffic simulation and congestion metrics.

Chapter 3

Background

3.1 Walker-Delta Constellations

A Walker-delta constellation is parameterised by the triplet $N/P/F$ (satellites / planes / phasing factor). Starlink’s operational shells use inclinations of $\sim 43^\circ$ and $\sim 53^\circ$, arranged in evenly spaced orbital planes with ± 2 intra-plane and ± 2 inter-plane inter-satellite links (ISLs). The regular lane structure provides coverage efficiency but creates angular anisotropy that affects geographic routing.

3.2 Routing Approaches

Greedy geographic routing selects the next relay based on local geometric progress toward the destination. It requires no global topology knowledge, but can fail in *routing voids* where no local relay satisfies the forwarding constraints.

Dijkstra shortest-path routing computes the globally optimal path with full topology knowledge. It cannot get stuck in voids; the trade-off is the need for a distributed link-state protocol (e.g., OSPF-TE) to maintain topology information.

3.3 The BPP Analytical Model

Wang et al. [1] model each orbital tier as a Binomial Point Process— N satellites placed independently and uniformly at random on a sphere. They derive the probability that a valid relay exists at each hop given the routing constraints (direction, dome angle, distance, and line-of-sight), then chain this across multiple hops via a Markov-chain formulation to obtain the end-to-end interruption probability. The model is mathematically tractable and has been validated against Monte Carlo BPP simulation, but not against real Walker-delta TLE data.

Chapter 4

Mathematical Model

4.1 Geometry

Distance between nodes u and v with ECEF positions $\mathbf{p}_u, \mathbf{p}_v$:

$$d(u, v) = \|\mathbf{p}_u - \mathbf{p}_v\|_2 \quad (4.1)$$

Propagation delay over link (u, v) :

$$\tau_{\text{prop}}(u, v) = \frac{d(u, v)}{c}, \quad c = 299,792.458 \text{ km/s} \quad (4.2)$$

Dome angle (geocentric angle between two nodes):

$$\theta_{\text{dome}}(u, v) = \arccos(\hat{\mathbf{p}}_u \cdot \hat{\mathbf{p}}_v), \quad \hat{\mathbf{p}} = \mathbf{p} / \|\mathbf{p}\| \quad (4.3)$$

Direction angle from current node c toward candidate v , relative to destination d :

$$\theta_{\text{dir}}(c, v, d) = \arccos\left(\frac{\mathbf{p}_v - \mathbf{p}_c}{\|\mathbf{p}_v - \mathbf{p}_c\|} \cdot \frac{\mathbf{p}_d - \mathbf{p}_c}{\|\mathbf{p}_d - \mathbf{p}_c\|}\right) \quad (4.4)$$

Line-of-sight condition (segment uv does not intersect Earth of radius $R_E = 6,378.137 \text{ km}$):

$$\min_{t \in [0, 1]} \|\mathbf{p}_u + t(\mathbf{p}_v - \mathbf{p}_u)\|^2 \geq R_E^2 \quad (4.5)$$

4.2 BPP Routing Constraints

The Wang/BPP algorithm applies three constraints to filter candidate relays v at each hop from current node c toward destination d :

$$c_1 : \theta_{\text{dir}}(c, v, d) \leq \theta_r \quad (4.6)$$

$$c_2 : \theta_{\text{dome}}(c, v) \geq \theta_s \quad (4.7)$$

$$c_3 : d(c, v) \leq d_{\text{th}} \text{ and LoS}(c, v) \quad (4.8)$$

c_1 (direction, [1] Sec. III-A) restricts the search to a forward cone; c_2 (dome angle) ensures geometric progress; c_3 (distance + LoS) limits hop length and requires clear sky. The relay minimising $\theta_{\text{dome}}(v, d)$ among valid candidates is chosen:

$$v^* = \arg \min_{v \in \mathcal{C}} \theta_{\text{dome}}(v, d) \quad (4.9)$$

If $\mathcal{C} = \emptyset$, the packet is **interrupted**.

4.3 BPP Analytical Interruption Probability

For tier k with N_k randomly placed satellites, the probability that no valid relay exists in that tier is:

$$q_k = (1 - p_k)^{N_k} \quad (4.10)$$

where $p_k = \Omega_{\text{valid}}^{(k)} / (4\pi R_k^2)$ is the fraction of the sphere surface satisfying all three constraints. The single-hop interruption probability over K tiers is:

$$P_{\text{int}}^{(1)} = \prod_{k=1}^K (1 - p_k)^{N_k} \quad (4.11)$$

Multi-hop interruption is derived via a Markov chain over dome-angle states [1]:

$$P_{\text{int}}^{(\text{multi})} = 1 - [\mathbf{T}^{\bar{H}}]_{\text{src} \rightarrow \text{dst}} \quad (4.12)$$

where $\mathbf{T}$ is the hop transition matrix and $\bar{H}$ is the expected hop count.

4.4 Dijkstra Path Cost

$$C(\pi^*) = \min_{\pi \in \Pi(s, d)} \sum_{i=0}^{H-1} \tau_{\text{prop}}(v_i, v_{i+1}) \quad (4.13)$$

4.5 Delay Model

Serialisation delay for packet of L bytes on link of capacity R bps:

$$T_s = \frac{8L}{R} \quad (4.14)$$

M/D/1 queuing delay (Pollaczek–Khinchine, deterministic service):

$$W_q = \frac{\rho}{2(1-\rho)} T_s, \quad \rho = \frac{\text{offered load}}{R} \quad (4.15)$$

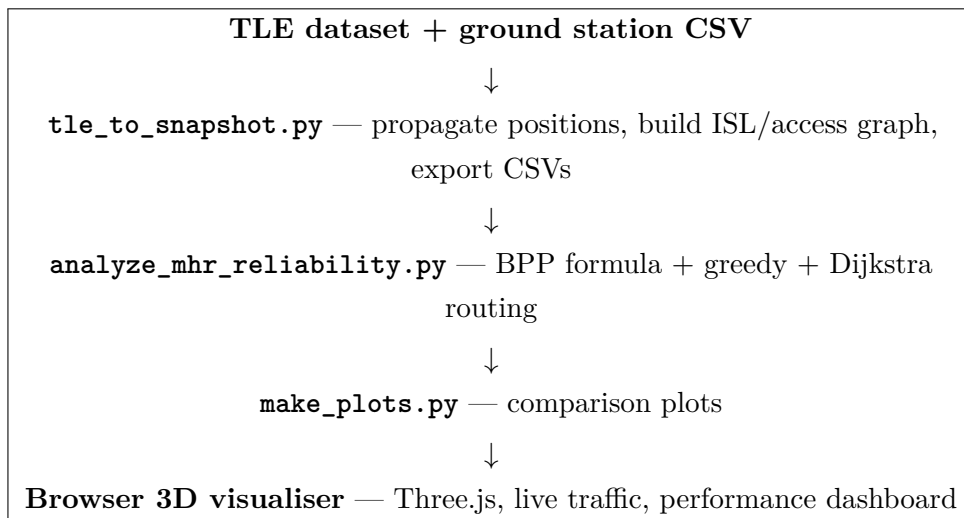
Total end-to-end delay:

$$\tau_{\text{E2E}} = \sum_{i=1}^H \left(\tau_{\text{prop}}^{(i)} + T_s + W_q^{(i)} \right) \quad (4.16)$$

Chapter 5

System Overview and Implementation

5.1 Pipeline



5.2 Dataset

The TLE file (`starlink.tle`) contains 9,887 raw records. After filtering for operational satellites (altitude 400–600 km, eccentricity < 0.01), 8,691 valid TLEs remain. Shell selection retains the two largest well-separated orbital groups: 2,670 satellites at $\sim 43^\circ$ inclination and 1,410 at $\sim 53^\circ$, totalling 4,080 satellites. The ground station CSV contains 250 real Starlink gateway locations.

Ten frozen-time snapshots are generated at 9-minute intervals starting at 21:30 UTC, 21 March 2026, using the Skyfield SGP4 propagator.

Table 5.1: Mean topology statistics across 10 epochs.

Quantity	Value
Satellites (2 shells)	4,080
Gateways	250
Total nodes	4,330
ISL edges per epoch	$\approx 7,850$
Access edges per epoch	$\approx 29,600$
Total edges per epoch	$\approx 37,450$
Mean ISL length	$\approx 1,246$ km
Mean access link length	$\approx 1,367$ km
Largest connected component	$\geq 99.6\%$ of nodes

ISL policy: intra-plane ± 2 neighbours, inter-plane ± 2 adjacent planes, max length 5,000 km; ISLs above $|\text{lat}| > 70^\circ$ are disabled (polar seam avoidance).

5.3 Routing Algorithms

5.3.1 Dijkstra

Standard binary min-heap Dijkstra on the ISL/access graph with propagation delay as edge weight. Gateways are endpoints only—they cannot serve as relay nodes mid-path.

5.3.2 BPP Greedy (Algorithm 1)

At each hop, all N nodes are scanned. Constraints c_1 , c_2 , c_3 (Eqs. 4.6–4.8) are applied. The candidate with smallest dome angle to destination is chosen. If none exists: interrupted. This matches the paper’s specification exactly, including relaxing angular constraints for gateway–satellite hops.

Algorithm 1: BPP Greedy Multi-Hop Routing**Input:** Source s , destination d , $(\theta_r, \theta_s, d_{th})$ **Output:** Path π , outcome

```

1  $c \leftarrow s$ ;
2  $\pi \leftarrow [s]$ ;
3  $visited \leftarrow \{s\}$ 
4 for  $hop = 1$  to  $MaxHops$  do
5   if  $d(c, d) \leq d_{th}$  and  $LoS(c, d)$  then append  $d$ ; return success
6    $\mathcal{C} \leftarrow \{v \notin visited : c_1 \wedge c_2 \wedge c_3\}$ 
7   if  $\mathcal{C} = \emptyset$  then return interrupted
8    $v^* \leftarrow \arg \min_{v \in \mathcal{C}} \theta_{dome}(v, d)$ ;
9    $\pi.append(v^*)$ ;
10   $visited.add(v^*)$ ;
11   $c \leftarrow v^*$ 
12 return interrupted

```

5.4 3D Visualiser

The browser visualiser uses Three.js (WebGL) with:

- **Instanced rendering:** 4,080 satellites in a single `InstancedMesh` draw call; $\sim 37,000$ edges in a single `LineSegments` draw call with per-vertex colours.
- **Animated packet:** a glowing sphere traversing the route hop by hop.
- **Background traffic:** four intensity presets (Low, Medium, High, Stress) generating continuous random gateway-to-gateway flows.
- **Live dashboard:** EMA utilisation ρ_e , M/D/1 queuing delay, packet loss, goodput, jitter, and a per-link congestion heatmap.
- **Dijkstra vs BPP comparison panel:** sends both routing algorithms sequentially and shows results side by side.

Gateway access links use a lower capacity (25 Mbps default) than ISL trunks (100 Mbps), making gateways visible bottlenecks in the congestion heatmap.

Table 5.2: Background traffic preset parameters.

Intensity	Rate (pkt/s)	Demand mult.	Max flows	Target ρ_{hot}
Low	8	400	80	< 0.25
Medium	35	1,500	220	0.40–0.60
High	80	900	320	0.70–0.90
Stress	160	2,800	500	$\rightarrow 1.0$

Chapter 6

Experiment Setup

- **Constellation:** full Walker-delta (4,080 satellites + 250 gateways).
- **Epochs:** 10 frozen-time snapshots, 9-minute steps (81 minutes total).
- **Pairs:** 300 random gateway-to-gateway source-destination pairs per epoch.
- **d_{th} sweep:** 3,000 / 3,500 / 4,000 / 5,000 km.
- **Fixed parameters:** $\theta_r = 45^\circ$, $\theta_s = 15^\circ$.
- **Queuing:** 1 Gbps ISL, 200 Mbps offered load ($\rho = 0.2$), $L = 1,500$ bytes.
At this load, $T_s \approx 0.012$ ms and $W_q \approx 0.0015$ ms per hop, so propagation dominates.

Chapter 7

Results

7.1 Interruption Probability

Table 7.1: Interruption probability — mean over 10 epochs $\times$ 300 pairs.

d_{th} (km)	Greedy	BPP pred.	Dijkstra	Abs. error
3,000	0.8743	0.0160	0.0000	0.8584
3,500	0.8490	0.0144	0.0000	0.8346
4,000	0.7857	0.0144	0.0000	0.7712
5,000	0.7443	0.0144	0.0000	0.7299

At $d_{\text{th}} = 4,000$ km, BPP predicts only 1.44% interruption while the empirical greedy simulator measures 78.57%—an absolute prediction error of **77.12 percentage points**. Dijkstra achieves **zero interruptions** across all configurations.

7.2 Hop Count ($d_{\text{th}} = 4,000$ km)

Table 7.2: Mean hop count at $d_{\text{th}} = 4,000$ km (delivered routes only).

Algorithm	Mean hops
Greedy	8.11
Dijkstra	12.65

Dijkstra uses more hops than greedy (12.65 vs 8.11 mean) because it minimises total delay rather than hop count—shorter hops yield lower per-hop propagation. Greedy uses fewer hops when it succeeds, but succeeds only 21.4% of the time.

7.3 End-to-End Latency ($d_{\text{th}} = 4,000$ km)

Table 7.3: Mean E2E latency (ms) at $d_{\text{th}} = 4,000$ km ($\rho = 0.2$, 1,500-byte packets).

Algorithm	Mean latency (ms)
Greedy	69.87
Dijkstra	66.89

Dijkstra achieves lower mean latency (66.89 ms vs 69.87 ms). At $\rho = 0.2$ the queueing contribution is negligible (< 0.1 ms total); propagation dominates.

Chapter 8

Analysis: Why BPP Fails on Real Starlink Geometry

8.1 The Core Contradiction

Method	Interruption @ $d_{\text{th}} = 4,000 \text{ km}$
BPP analytical	0.0144
Greedy empirical	0.7857
Dijkstra	0.0000

The BPP formula is correct under its own assumptions: with 4,080 randomly placed satellites, a valid relay almost always exists somewhere in the required directional cone. The problem is that *real Starlink satellites are not random*.

8.2 Walker-Delta Creates Systematic Voids

Walker-delta orbital planes produce angular lanes with regular gaps between them. If the routing direction from the current node to the destination points across an inter-lane gap, no satellite satisfies c_1 within d_{th} —even though the destination is reachable globally via a different path. The BPP formula averages over isotropic placements and cannot predict these systematic directional voids.

The result is stark: BPP sees 4,080 satellites at high density and predicts trivially easy routing. The greedy algorithm, working in actual Walker-delta geometry, fails 78.57% of the time.

8.3 Dijkstra as a Diagnostic

The fact that Dijkstra achieves zero interruptions is crucial:

- The Walker-delta graph is globally well-connected ($> 99.6\%$ in a single component).
- Every tested pair has a valid route—just not one reachable by local greedy forwarding.

- Routing failures are entirely due to the algorithm, not to coverage gaps.

This confirms the engineering implication: real LEO networks require link-state routing with global topology knowledge, not geographic-greedy forwarding.

8.4 BPP Error Across d_{th}

Table 8.1: BPP absolute prediction error — full Walker-delta constellation.

d_{th} (km)	BPP pred.	Greedy emp.	Abs. error (pp)
3,000	0.0160	0.8743	85.84
3,500	0.0144	0.8490	83.46
4,000	0.0144	0.7857	77.12
5,000	0.0144	0.7443	72.99

Across all d_{th} values, BPP underestimates greedy interruption by 73–86 percentage points ($> 98\%$ relative error in all cases).

Chapter 9

Discussion, Limitations, and Future Work

9.1 Discussion

1. **Validate before applying.** A model that works under random-placement assumptions may be arbitrarily wrong for structured constellations. BPP should not be used for quantitative routing-reliability predictions on Walker-delta topologies without empirical validation.
2. **Algorithm and topology interact.** The BPP algorithm was designed for BPP topology. Applied to Walker-delta geometry it combines algorithmic limitations (local-only, no backtracking) with structural voids, producing 78.57% interruption where BPP predicts 1.44%.
3. **Topology-aware routing is essential.** Dijkstra’s zero interruption rate shows that the same satellite mesh supports reliable routing when global topology knowledge is available. This strongly motivates link-state protocols for operational LEO networks.

9.2 Limitations

- Real Starlink routing, traffic, and handover policies are proprietary.
- Snapshots are frozen in time; real topology changes continuously.
- ISL assignment is a geometric heuristic, not SpaceX’s actual policy.
- Gateway scheduling and RF interference are not modelled.
- The visualiser uses a simplified, calibrated traffic model—not a Starlink SLA predictor.
- Dijkstra is a baseline; it is not a claim about SpaceX’s actual protocol.

9.3 Future Work

- Realistic traffic matrices (gravity models from population density).
- Additional routing algorithms (GPSR void avoidance, segment routing).
- Handover and link-failure modelling.

-
- Full parameter sensitivity sweep (θ_r , θ_s , all Starlink shells).
 - NS-3 validation (a stub C++ scenario is already included).
 - High-load batch experiments to quantify joint queueing + interruption effects.

Chapter 10

Conclusion

This project built a complete Starlink-like multi-hop LEO routing simulator from real TLE data. The simulator generates multi-epoch network snapshots, implements both the Wang/BPP greedy algorithm and Dijkstra routing, and computes the closed-form BPP analytical prediction on the same topology.

The central finding is stark. On the full 4,080-satellite Walker-delta constellation at $d_{\text{th}} = 4,000$ km:

BPP analytical prediction	1.44% interruption
Empirical greedy routing	78.57% interruption
Absolute prediction error	77.12 percentage points
Dijkstra interruption	0.00%

The BPP model is mathematically valid under its random-placement assumption. It fails on real Starlink geometry because Walker-delta orbital lanes create systematic directional voids that make greedy routing fail even where global routing succeeds. The zero Dijkstra interruption rate confirms that the issue is the routing algorithm, not the topology.

The practical lesson is clear: real LEO networks need link-state routing with full topology knowledge. Geography-based greedy forwarding, as formalised in the BPP model, is unreliable on structured Walker-delta constellations regardless of satellite density.

The 3D visualiser makes these dynamics directly observable, showing congestion, routing voids, and the contrast between Dijkstra paths and failed greedy routes in real time.

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